

Zarin Tasnim

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PROFESSIONAL SUMMARY

– Master’s graduate in Computer Science with strong foundations in software engineering, algorithms, and system design, experienced in developing scalable client-server backend applications using Java and object-oriented principles within Agile environments.

TECHNICAL SKILLS

Programming Languages: Java, SQL, Python, JavaScript

Backend Development: RESTful API design, client-server architecture, object-oriented design, application performance optimization

Frameworks: Spring Boot, Hibernate, Django

Databases: MySQL, PostgreSQL, SQLite; relational schema design, complex queries, indexing, transaction management

Systems: Linux/UNIX environments, HTTP protocol, web servers

Tools: Git/GitHub, IntelliJ IDEA, Docker

Development Practices: Agile/Scrum, Unit Testing (JUnit), debugging, code reviews, complexity analysis

EXPERIENCE

Customer Support Ticket Management System | Backend Developer (Self Project) **Dec 2025 – Feb 2026**

Toronto, Ontario

- Designed and implemented a scalable, modular client-server backend application using Java/Spring principles.
- Developed and optimized RESTful APIs and relational schemas using PostgreSQL, ensuring transactional integrity and efficient query performance.
- Optimized database queries and indexing strategies to improve response time under concurrent usage.
- Applied clean architecture, input validation, and unit testing to deliver maintainable, production-ready code.

Backend Scheduler System | Backend Developer (Self Project) **Nov 2025 – Dec 2025**

Toronto, Ontario

- Built a scalable client-server backend system handling concurrent scheduling requests with structured validation and persistence logic.
- Designed modular service layers with clear separation of concerns and reusable components.
- Implemented structured logging, centralized exception handling, and performance tuning to ensure reliability in production-like environments.

Backend System for Component Discovery | Masters Thesis **Sept 2023 – Jun 2025**

Concordia University, Montreal, QC

- Architected and implemented a modular backend application using **Java and Python**, applying service-oriented design principles.
- Designed and optimized data retrieval and ranking mechanisms, improving system response efficiency for complex query operations.
- Applied object-oriented design and modular service layering to maintain clean, extensible backend architecture.
- Delivered tested and documented production-quality code; results published in peer-reviewed IEEE conference.

Research Assistant **Sept 2021 – Jun 2024**

Concordia University, Montreal, QC

- Designed and maintained backend services in Java and Python, implementing structured APIs, validation logic, and data-processing workflows.
- Debugged and optimized algorithms to ensure performance stability and reliability.
- Collaborated in cross-functional teams, contributing to documentation, testing, and technical reviews.

Simulated Distributed Client/Server Application **Sept 2021 – Dec 2021**

Concordia University, Montreal, QC

- Designed and implemented a distributed client/server system communicating over **RESTful HTTP/JSON APIs**.
- Developed backend components in Python and JavaScript with structured request handling and fault-tolerant logic.
- Simulated real-time request/response workflows and measured latency to validate system reliability under network constraints.

EDUCATION

Master of Science in Computer Science | Concordia University, Montreal, QC **Sept 2021 – Jun 2025**

Bachelor of Science in Computer Science | BRAC University, Dhaka, Bangladesh **Sept 2016 – Dec 2020**